Energy Price Prediction and Optimisation Model for Dynamic-Rate Energy Contracts in Belgium

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1 Introduction: The Problem

The transition to renewable energy sources has fundamentally altered the behaviour of wholesale electricity markets, such as the Belgian EPEX SPOT market. Because generation sources like wind and solar are weather-dependent and non-dispatchable, energy grids frequently experience periods of severe oversupply or acute scarcity. This results in extreme price volatility, where the cost per megawatt-hour (MWh) can plunge into negative territory during sunny, windy weekends, or violently spike beyond €500/MWh during cold, still winter evenings.

For a consumer or facility with a dynamic energy contract and flexible energy-intensive tasks (such as EV charging, heavy HVAC loads, industrial manufacturing, or battery storage), this volatility represents an arbitrage opportunity: purchase, store and consume energy when it is abundant and cheap, and avoid consumption (or discharge stored energy) when prices surge.

However, predicting these optimal operating windows is an interesting forecasting problem. Standard heuristic triggers (e.g., "always run tasks at 2:00 AM" or "activate if the price is below €40") suffer from two fatal flaws:

- **Opportunity Loss:** They fail to adapt to seasonal shifts and weather patterns, often missing deeply negative pricing events.
- **Asymmetric Risk (Fat-Tails):** When a simple rule fails during a period of grid stress, the resulting penalty is not average; the consumer is exposed to catastrophic price spikes that wipe out weeks of accumulated savings.

To operate profitably, a scheduling model requires a controller that can accurately forecast market conditions up to seven days in advance, while quantifying the statistical risk of being wrong.

2 Energy Market Landscape and Feature Selection Rationale

2.1 Grid Operations and the 15-Minute Wholesale Market

The fundamental challenge of modern power grids is the physical requirement that electricity supply and demand must remain perfectly balanced at all times. Historically, this was achieved by dispatching controllable fossil fuel and nuclear plants to meet predictable human demand profiles. However, the aggressive integration of intermittent renewable energy sources, specifically wind and solar, has inverted this paradigm. The grid is now dictated by weather volatility, leading to frequent periods of severe oversupply (resulting in zero or negative prices) and sudden supply deficits (causing aggressive price spikes).

To accurately price these rapid fluctuations, wholesale energy exchanges, such as the Belgian EPEX SPOT market, operate using discrete time blocks. While historically cleared on an hourly basis, the market has transitioned toward a 15-minute resolution. This granular 15-minute settlement period is necessary to account for sudden meteorological shifts, such as a cloud cover moving over a major solar installation or a sudden drop in offshore wind speeds. The clearing price for each 15-minute block is determined by the "merit order effect," where the most expensive generation unit required to meet demand sets the price for all market participants in that block.

2.2 The Mechanics of Dynamic Energy Contracts

Traditionally, end-users were insulated from wholesale market volatility through fixed-rate energy contracts, where energy suppliers absorbed the risk of price fluctuations. A *Dynamic Energy Contract* fundamentally alters this relationship by directly passing the wholesale day-ahead (or intraday) market prices through to the consumer, typically with a minor fixed markup and volumetric taxes.

Under a dynamic contract, the cost of electricity changes every 15 minutes based on the EPEX SPOT clearing prices. This creates a financial ecosystem where demand flexibility is highly rewarded. Consumers or automated systems that can defer energy-intensive tasks (e.g., EV charging, industrial heating, water pumping) to periods of renewable oversupply can achieve near-zero or even negative energy costs. The objective of the predictive model is to exploit this exact mechanism, acting as an automated scheduling agent that mathematically guarantees task execution during the most statistically favorable 15-minute blocks.

3 Preliminary Analysis of Energy Price Data

3.1 Empirical Distribution of Wholesale Prices

A fundamental architectural decision in this model is the formulation of the objective function as a binary classification task rather than a continuous price regression. This decision is strictly empirical, driven by an exploratory data analysis of the historical Belgian EPEX SPOT price distribution.

When visualising the probability density of wholesale electricity prices over the target period (see Figure 1 below), the data does not follow a standard normal distribution. Instead, it exhibits a distinct bimodal distribution representing two entirely different physical states of the power grid.

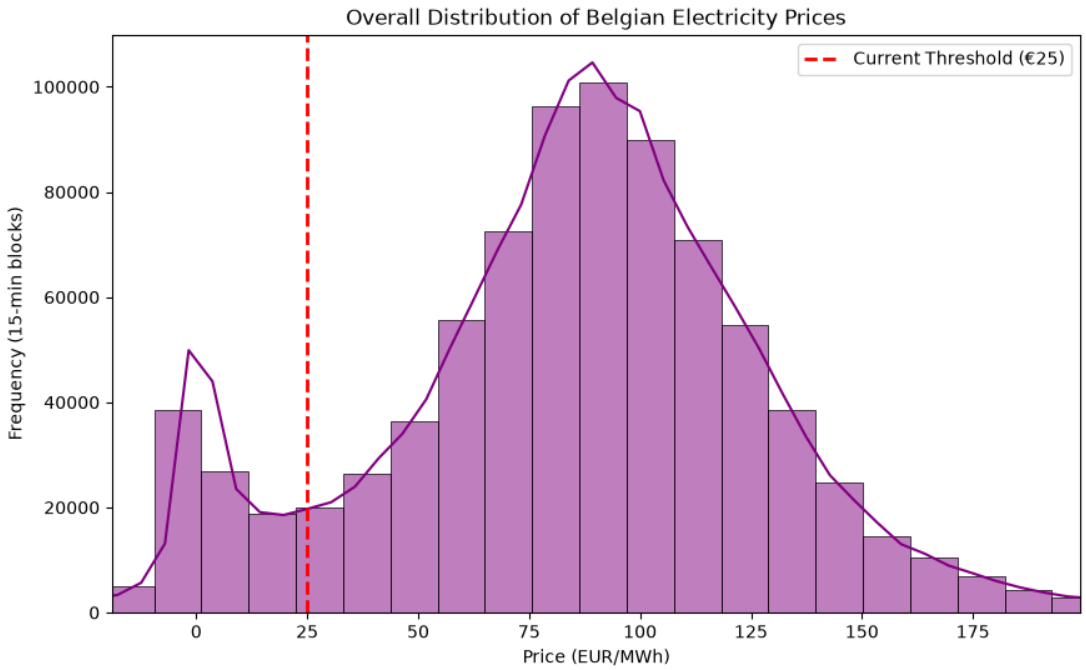


Figure 1: Visual representation of the bimodal distribution of energy prices.

The primary distribution is a bell curve centred roughly around €85/MWh. This peak represents standard grid operations where traditional merit-order economics apply, and demand is met by a steady stream of energy supply through, fossil fuel, nuclear, and potentially some renewable energy.

However, there is a pronounced secondary peak clustering near €0/MWh (and frequently dipping into negative prices). This secondary distribution represents grid anomalies: periods of severe renewable energy oversupply, typically triggered by high offshore wind yields or peak solar irradiance coinciding with low baseline demand.

3.2 The €25/MWh Threshold

Between these two distinct pricing regimes exists a natural "valley" or local minimum in the frequency distribution. The trough that separates the standard normal grid condition peak from the cluster of low energy prices sits right around the €25/MWh mark.

Setting the classification threshold at $\leq$€25/MWh allows the model to cleanly slice the dataset at this empirical boundary. This implies the model does not rely on an arbitrary definition of "cheap" energy, and instead the €25 threshold isolates the exact subset of hours where the grid operates at a condition of surplus energy.

3.3 Classification over Regression

Attempting to forecast these prices using a continuous regression model introduces significant operational hazards and hassles. Regression algorithms inherently optimise for loss functions like Mean Squared Error. When faced with a bimodal distribution, a regressor might predict a "safe" mathematical mean between the two peaks (potentially outputting values around €40 to €50/MWh) to minimise penalty.

For an automated scheduling agent, predicting a vague middle-ground price is practically useless. The battery or energy-intensive task does not need to know if the price will be exactly €8.50 or -€2.10; it only needs a high-confidence probabilistic signal that the grid is entering the secondary, low-cost regime. By discretising the target variable into a binary "Cheap" ($\leq$€25) versus "Expensive" ($>$€25) framework, we force the decision trees to identify the physical triggers that cause a shift between these two grid states.

3.4 Seasonal and Hourly Variation

Here we highlight the failure of standard heuristic triggers, such as static time-based rules, to consistently capture low-cost opportunities while avoiding severe financial penalties. This failure stems from two interconnected structural phenomena: macro-scale seasonal divergence and micro-scale intraday volatility.

Firstly, the probability density of wholesale electricity prices changes dramatically across seasons due to varying meteorological conditions and human demand profiles.

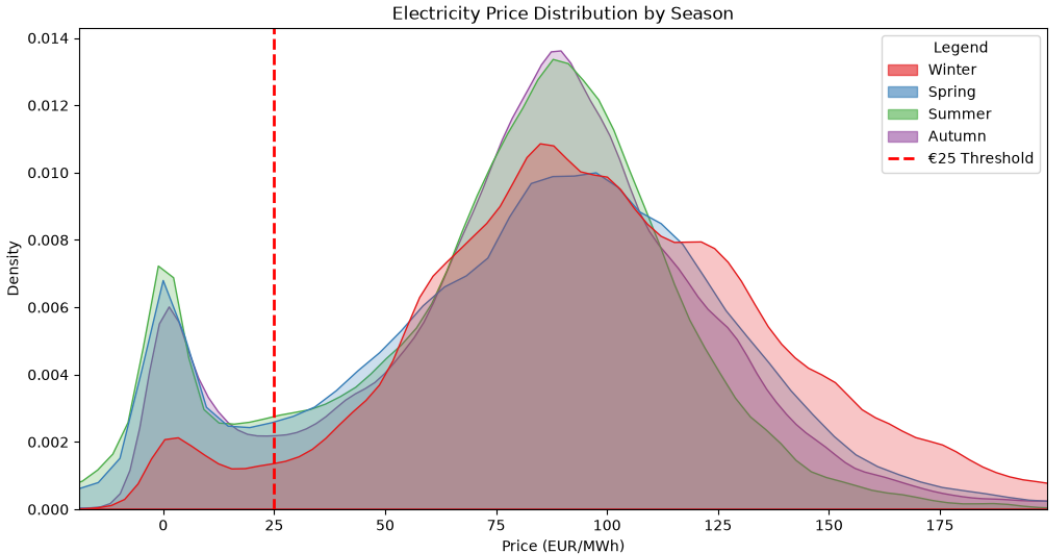


Figure 2: Visual representation of the seasonal variation in distribution of energy prices.

Evaluating the seasonal density plot (Figure 2) offers insight into general rules governing the price distribution during various seasons. Firstly, there is a clear bimodal separation in summer, autumn and spring. The density curves for summer (green), autum (purple), and spring (blue) exhibit a prominent secondary peak centred near €0/MWh, reaching probability densities of approximately 0.0072, 0.0060 and 0.0068 respectively. A substantial proportion of total probability mass sits strictly to the left of the red dashed €25/MWh threshold line. This confirms that deep price collapses ($\leq$€25/MWh) are heavily concentrated in high-renewable seasons.

Secondly, in stark contrast, there is a notable suppression of this second low-cost peak in winter. The winter distribution (red) flattens dramatically below the €25/MWh threshold, reaching a meager density peak of barely 0.002 near €0/MWh. Cheap hours ($\leq$€25/MWh) become sporadic during winter rather than predictable daily occurrences.

Lastly, the right tail of the price distribution in winter is considerably nore bloated in comparison to the other seasons. While summer, autumn, and spring density tails decay rapidly above €120/MWh, the winter curve features a broad, heavily bloated right tail extending past €175-€200/MWh.

Furthermore and even more critically, analysing price distribution within specific 15-minute hourly blocks (Figure 3) reveals high structural variance.

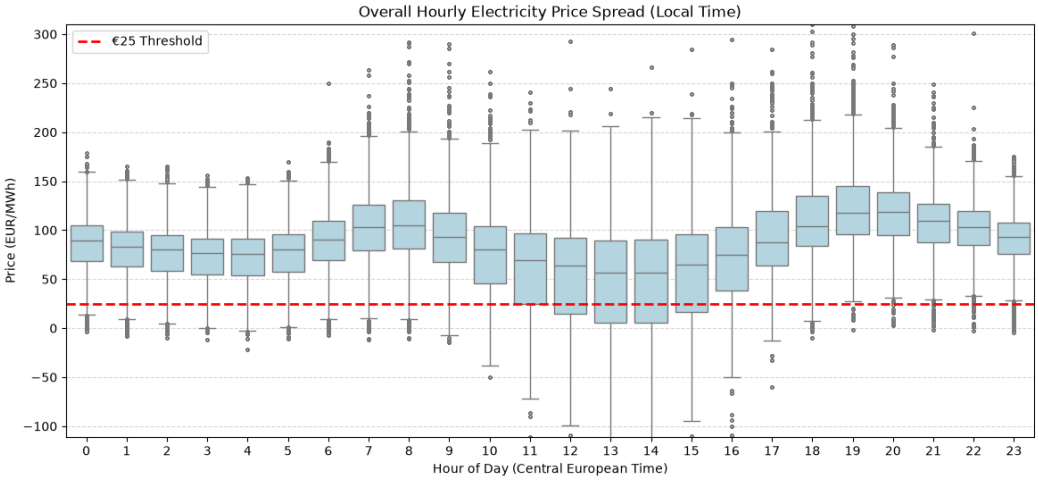


Figure 3: Visual representation of overall intraday distribution of energy prices.

While the median prices during the early afternoon may sit the lowest out of all hours of the day, they are also the most prone to the most violent price spikes. The distribution exhibits severe right-skewed fat tails. Cloud coverage over concentrated solar arrays and drop-offs in North Sea offshore wind generation can cause prices within an otherwise "cheap" 15-minute window to abruptly spike to several hundred Euros per MWh.

In this way, a naive agent relying purely on historical time window averages will inevitably execute trades during these anomalous spike events, incurring financial losses that eliminate previously accumulated operational savings.

To corroborate all claims made further, an illustration of the median price for each hour of the day per season (Figure 4) can be made.

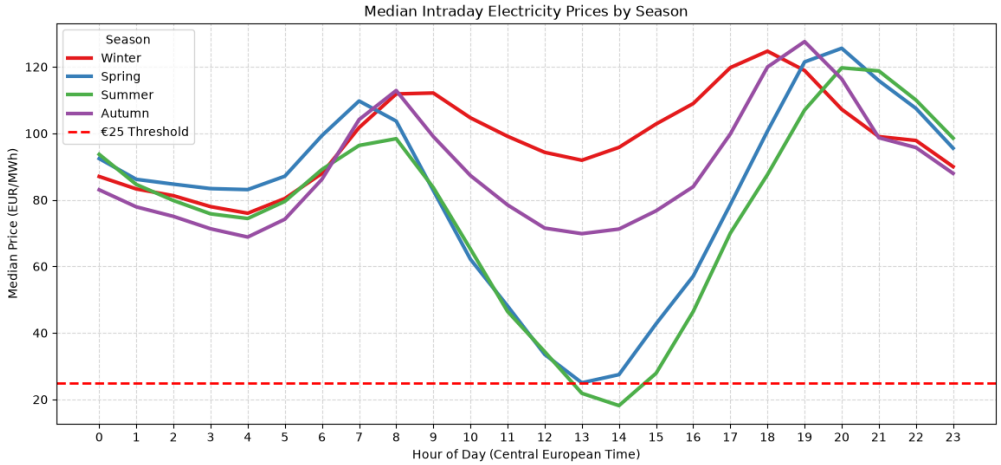


Figure 4: Visual representation of median intraday energy prices per season.

Figure 4 further shows that electricity prices changes dramatically across seasons due to varying meteorological conditions and human demand profiles. In summer and spring midday prices experience extreme downward pressure, with median prices hovering around the €25/MWh threshold. This is driven by high renewable energy production (specificallt solar) yields coinciding with moderate baseline load, producing the characteristic "duck curve."

Whereas in autumn and winter, increased heating demand and suppressed solar irradiance flatten the intraday curve. Midday prices remain anchored near the €70–€100/MWh baseline peak set by marginal fossil fuel generators, rendering midday operational rules ineffective.

4 Model Architecture and Objective

4.1 Feature Space Formulation

To provide the gradient boosting classifier with a comprehensive representation of physical grid dynamics, market momentum, and weather variability, raw data streams were integrated from multiple domain sources:

- **Wholesale Market and Gas Prices:** EPEX SPOT 15-minute day-ahead electricity settlement prices and Title Transfer Facility (TTF) natural gas benchmark futures.
- **Exogenous Meteorological Telemetry:** High-resolution Open-Meteo weather reanalysis and forecast data across multiple spatial nodes (Brussels, Maasmechelen, and North Sea offshore grid points), capturing wind speeds, ambient temperatures, and Global Tilted Radiation (GTI) tuned to a 35° panel angle.
- **Grid Infrastructure and Generation Metrics:** ENTSO-E telemetry (ods datasets) detailing load demand, cross-border Net Transfer Capacity (NTC import/export capacities with the UK), implicit net position, and fuel-specific generation streams (fossil gas, nuclear, onshore wind, offshore wind, and solar PV).

To transform static point-in-time observations into predictive indicators, an automated feature engineering engine computes a suite of temporal transformations across every fundemental variable:

- **Autoregressive Lags ($t-1$, $t-2$, $t-7$):** Capturing diurnal (96 steps), 2-day (192 steps), and weekly (672 steps) seasonal memory.
- **Rolling Moving Averages (3D, 7D, 14D):** Smooth moving averages computed over 288, 672, and 1344 15-minute windows to establish baseline price and generation trends.

4.2 Probabilistic Discretisation Framework

The output of the prediction model is the probability it computes of the price falling below €25. However, this percentage by itself is not very telling as to whether it is better to consume energy during a time when the price is known or for the time the model gave the prediction, as the percentage does not tell us anything about the known price of electricity. To answer this question, we need to construct a model which relates the known price of electricity to a potential price of electricity given the prediction output of the model.

Rather than treating the XGBoost output as a hard binary classification decision using a fixed cutoff threshold (e.g., $\hat{y} = 1$ if $p > 0.5$), the decision engine evaluates the model's raw output probability $\hat{p_i} = P(y_i = 0 \mid \mathbf{x}_i)$, representing the estimated likelihood that the clearing price for period i will fall below or equal to the €25/MWh threshold given market state $\mathbf{x}_i$.

To systematically evaluate whether these predicted probabilities reflect true real-world frequencies, the continuous probability range $[0, 1]$ is partitioned into 10 uniform probability intervals (bins):

Definition 4.2.1 Let

$$B_1 = \left[0, \frac{1}{10}\right] \text{ and } B_k = \left(\frac{k-1}{10}, \frac{k}{10}\right], \text{ for } k \in \{2, ..., 10\}$$

denote the 10 probability bins partitioning the set of outputs the prediction model gives.

For each probability interval B_k, two key statistical quantities are computed over the out-of-sample test set:

Definition 4.2.2 Define the Empirical Accuracy ($\overline{y}_k$) as the actual fraction of time blocks within bin B_k where the true market price was observed to be $\leq$€25/MWh, given by

$$\overline{y}_k = \frac{1}{|B_k|} \sum_{i \in B_k} \mathbb{I}(y_i = 0)$$

where $\mathbb{I}$ is the indicator function equal to 1 if the empirical condition $(y_i = 0)$ holds, and 0 otherwise.

Definition 4.2.3 Define the Mean Predicted Probability ($\overline{p}_k$) as the true arithmetic mean of the model's output probabilities assigned to samples within bin B_k, given by

$$\overline{p}_k = \frac{1}{|B_k|} \sum_{i \in B_k} \hat{p}_i$$

We also consider the set of empirical accuracy values for each probability bin:

Definition 4.2.4 Let A denote the set of empirical values for all probabily bins, given by

$$A = \{\overline{y}_k : 1 \leq k \leq 10\}.$$

4.3 Empirical Calibration vs. Midpoint Distortion

Because tree-based ensemble classifiers tend to produce non-uniform probability distributions, frequently clustering predictions densely near extreme bounds (e.g., 1–2% or 97–98%), the geometric midpoint of bin k fails to reflect the true expected value of the model's predictions within that bucket.

Comparing $\bar{t}_k$ directly to $\overline{p}_k$ eliminates this distribution-induced skew. A perfectly calibrated model satisfies

$$\overline{y}_k = \overline{p}_k \text{ for all } k \in \{1, 2, ..., 10\}$$

Visualising $\overline{y}_k$ against $\overline{p}_k$ produces an unskewed reliability diagram (Figure 5), revealing exact regions of model over-confidence ($\overline{y}_k > \overline{p}_k$) or under-confidence ($\overline{y}_k < \overline{p}_k$).

4.4 Integration into Downstream Model Mechanics

Quantifying bin-level empirical performance serves a critical structural purpose: it isolates downstream execution logic from potential model miscalibration.

Rather than feeding raw model probabilities $\hat{p}_i$ directly into the scheduling algorithm under the assumption of perfect calibration, the execution engine maps predictions into the empirical lookup table. For any real-time prediction falling within interval B_k, the engine retrieves the empirical historical success rate $\overline{y}_k$ and its associated conditional loss distributions (these will be introduced briefly).

This transforms the probabilistic ML output into an empirically grounded risk metric, ensuring that the Expected Value calculation operates on verified historical outcome frequencies rather than raw confidence scores.

4.5 The Quantitative Objective

The goal of this model is to minimise the cost of running an electrical appliance by selecting a time to run the appliance when the price of energy is predicted to be inexpensive. One way of doing this is by maximising the relative savings described by the difference in the cost of running the electrical appliance at the known price of energy (which is now referred to as the default alternative) and the predicted price of energy.

Before we can describe such a function we have to develop an understanding of the environment. We begin with the following definitions.

Definition 4.5.1 The following are definitions for important environment variables.

- Let E denote the energy usage of the appliance in MW during a single 15-minute block.
- Let P_t denote the price of energy at block t.
- Let D denote the set of 15-minute blocks which make up the default alternative schedule.

- Let $P_{default}$ given by
$$P_{default} = \frac{1}{|D|} \sum_{i \in D} p_i$$
denote the average price of energy during the default alternative.

We are now ready to define the relative savings function.

Definition 4.5.2 Let S_t given by
$$S_t = E(P_{default} - P_t)$$
be the cost difference between runnning the applience during the default alternative and time block t.

This function produces two notable resutls:

- $S_{win} \in \{S_t : S_t > 0\}$
- $S_{loss} \in \{S_t : S_t \leq 0\}$

Here, S_{win} simply refers to if delaying the operation saved the consumer money, and S_{loss} if delaying the operation lost the consumer money. These two results can be related to form the expected value of operating at time t.